ADAPT: Adaptive Dual-projection Architecture for Perceptive Traversal

Abstract— **Agile humanoid locomotion in complex 3D environments requires reasoning about both ground-support geometry and body-level traversable space under the strict latency constraints of high-frequency control. Existing perceptive locomotion methods often face two coupled limitations: compact 2.5D maps are efficient but miss vertical clearance constraints, whereas dense 3D representations provide richer geometry at higher computational cost; meanwhile, fixed sensing scale cannot adapt to changes in motion speed and local environmental complexity. We propose ADAPT (Adaptive Dual-projection Architecture for Perceptive Traversal), a perception-control framework that addresses both representation efficiency and sensing-scale adaptation. ADAPT decomposes local 3D geometry into two compact projections: a horizontal elevation map for terrain support and a vertical distance map for body-level clearance. It further treats the sensing radius of the vertical map as a reinforcement learning policy action, enabling the robot to expand its perceptual horizon for long-range anticipation and contract it for higher local resolution in cluttered spaces. Experiments show that ADAPT achieves a 94.75% success rate across diverse 3D traversal tasks, and reduces the exteroceptive observation input dimensionality by 98.6% relative to our implementation of the Gallant baseline, while also lowering deployment latency and computational resource usage. The learned policy achieves zero-shot transfer to a Unitree G1 humanoid robot and enables real-world space traversal.**

I. Introduction

Humanoid robots are expected to move through human-scale environments that contain stairs, cluttered passages, protruding structures and other three-dimensional constraints. For successful traversal, the controller depends on not only estimating where the feet can step on, but also understanding whether the whole body can safely pass through the surrounding space, including overhead and lateral obstacles that may cause collision or loss of balance. This makes perceptive traversal a coupled problem of terrain understanding, body-level clearance reasoning and real-time whole-body control.

However, onboard perception for legged robots is constrained by strict latency and bandwidth requirements. Existing representations therefore tend to emphasize either efficiency or geometric completeness. Conventional 2.5D elevation maps [1], [2], [3], [4], [5], [6], [7], [8], [9], [10], [11] provide compact terrain geometry for foothold reasoning, but they are intrinsically blind to overhangs, side walls and body-level clearance constraints. Depth-based representations [12], [13], [14], [15], [16], [17], [18], [19], [20] offer direct visual observations, yet they can be affected by view changes, occlusion and illumination variation. Volumetric approaches [21], [22], [23] encode richer 3D geometry, but their dimensionality and processing cost make high-frequency policy learning and deployment difficult. These limitations necessitate a representation that preserves 3D traversability while remaining computationally compact [24], [25], [26].

A second limitation is that perception is usually acquired with a fixed spatial range [21], [9], [4], which is misaligned with the changing information demand of locomotion: fast motion needs a longer horizon while cluttered traversal needs higher local resolution. A perceptive controller should therefore decide not only how to act given perception, but also how much spatial context should be sensed.

To address these two issues, this paper proposes **ADAPT** (**A**daptive **D**ual-projection **A**rchitecture for **P**erceptive **T**raversal) that restructures local 3D geometry into two orthogonal and information-dense projections. A horizontal elevation map describes ground geometry for stable foothold selection, while a vertical distance map describes body-level traversable space for avoiding overhangs and obstacles. Instead of using a fixed perceptual horizon, ADAPT further exposes the sensing radius of the vertical distance map to the reinforcement learning policy as a control variable. The horizontal elevation map retains a fixed spatial extent and sampling resolution. The policy can thus expand the vertical map's perceptual horizon during fast traversal and contract it for fine-grained local reasoning in cluttered scenes. In this way, ADAPT jointly addresses what spatial information should be represented and how much of it should be sensed at each moment.

Empirical evaluation shows that ADAPT achieves a mean success rate of 94.75% across eight terrain categories and reduces per-frame actor exteroceptive dimensionality by 98.6% relative to Gallant. Under matched training settings, it uses 56% less peak GPU memory and 32.7% of Gallant's iteration time. On the same local computer, perception-to-command-publication latency and sampled peak GPU memory decrease by 29.6% and 58.4%, respectively (Section IV).

The main contributions of this work are:

- Proposing a highly efficient formulation that decouples complex 3D environments into compact projections. This paradigm reduces observational dimensionality while rigorously preserving essential geometric constraints, thereby ensuring the high-frequency perception strictly required for agile locomotion.
- Introducing a bidirectional closed-loop mechanism that embeds perceptual configurations directly into the reinforcement learning action space. This formulation empowers the robot to autonomously modulate its sensory settings based on instantaneous dynamic states and environmental complexity, thereby achieving an optimal trade-off between navigational safety and precise spatial

perception.
- Demonstrating robust zero-shot sim-to-real transfer on a Unitree G1 humanoid robot with successful navigation through diverse structural challenges.

II. Related Works

A. Perception for Legged Locomotion

Table I compares elevation, image/depth, and volumetric or point-based representations in terms of geometry, input size, and sensing adaptation.

TABLE I: ADAPT comparison with existing works. Sensory dimension counts the per-frame perceptual input elements before encoding, excluding proprioception, observation history, and critic-only privileged inputs.

Method	Perceptual Representation	Native Sampling	Sensory Dimension	Overhead	Adaptive Sensing
PIM [4]	Elevation Map	0.10 m	96	✗	✗
Beamdojo [5]	Elevation Map	0.10 m	225	✗	✗
Vb-com [6]	Elevation Map	0.10 m	225	✗	✗
Parkour [14]	Depth Image	48×64 px	3072	✓	✗
Omni [23]	Point Cloud	0.05π rad	251	✓	✗
Gallant [21]	Voxel Grid	0.05 m	40960	✓	✗
HumanoidPF [27]	Joint Position	39 dims	39	✓	✗
ADAPT (ours)	Dual Projection	**0.025π rad**	**578**	✓	✓

Elevation maps [1], [2], [4], [5], [6], [7], [11] provide efficient 2.5D grids for foothold reasoning. However, their height-field assumption inherently discards vertical structures like overhangs and side walls, missing the body-level clearance constraints crucial for humanoid traversal in 3D clutter.

Depth and vision-based methods [13], [12], [14], [15], [16], [18], [19], [20] offer richer scene observations for agile locomotion. Yet, their unstructured inputs heavily burden the learning system and are susceptible to occlusions, viewpoint shifts, and illumination changes. While reduced visual encodings mitigate this, they tend to entangle ground geometry with visual noise rather than explicitly separating the needed spatial constraints.

Volumetric, point-cloud and multi-plane representations aim to recover more complete 3D spatial awareness [9], [21], [22], [23], [27]. These methods are better suited to collision avoidance and constrained-space traversal because they can encode obstacles beyond the terrain surface. Their main drawback is the cost of maintaining and processing dense 3D structure. Voxel grids scale cubically with resolution and incur high memory traffic [24], leading to severe inference latency (e.g., 4.4 Hz in [25]) that fails the ≥ 10 Hz requirement for closed-loop control [26] and inflated training iteration times (e.g., $>$ 22 s/iter in [21]). Point-based representations require careful sampling and aggregation to remain stable for policy learning. Multi-plane methods mitigate the cubic cost but leave vertical clearance implicit, forcing the policy to infer the gap between an overhead obstacle and the supporting ground across plane indices rather than from a direct distance measurement. The increased latency and observation dimensionality can slow training and reduce the responsiveness needed for high-speed humanoid motion.

ADAPT bridges these approaches with two compact projections that explicitly separate ground support from body-level clearance without encoding the full local volume.

B. Adaptive Perception and Behavior Modulation

The need for adaptive perception arises from the non-uniform information demand of locomotion: fast traversal requires a long perceptual horizon to anticipate obstacles, whereas cluttered traversal benefits more from high-resolution local geometry. A fixed sensing range cannot satisfy both simultaneously.

Spatial attention and foveation methods select informative regions or suppress irrelevant visual content [28], [29], while multi-range mapping assigns different densities to nearby and distant regions [30]. However, these mechanisms adapt perception after the sensing structure has been specified; the spatial extent and resolution levels remain fixed by design rather than continuously selected by the policy.

Active perception mechanisms change the sensing focus or viewpoint according to task context [31], and perceptive locomotion methods adapt gait and footholds based on terrain observations [14], [4], [5], [6]. Nevertheless, perception is typically organized as an upstream module with a fixed range, and the downstream policy adapts actions to those observations. The missing component is a feedback path in which the same policy that generates motion also regulates the amount of spatial information it receives.

ADAPT closes this feedback loop by making only the vertical map's sensing radius a policy action and including the previous radius in the observation. The horizontal map retains fixed coverage and resolution. Reinforcement learning jointly adapts locomotion and sensing range without changing the sensor pose.

III. Methods

A. Problem Formulation

The perceptive traversal task is formalized as a goal-conditioned Partially Observable Markov Decision Process (POMDP). At step t, policy π_θ receives the robot state and local geometry in $\mathbf{o}_t$ and outputs joint targets and a vertical-map sensing radius, $\mathbf{a}_t = (\mathbf{q}_t, r_t)$.

Driven by the stochastic transition dynamics $\mathcal{T}$, the agent seeks to learn an optimal policy π^* that maximizes the expected cumulative discounted return:

$$\pi^* = \arg\max_{\pi} \mathbb{E}_\pi \left[\sum_{t=0}^{T} \gamma^t \mathcal{R}(s_t, a_t) \right], \tag{1}$$

where $\gamma \in (0, 1)$ is the discount factor and T is the horizon. The scalar reward $\mathcal{R}(s_t, a_t)$ encodes multiple objectives, including velocity tracking, energy efficiency, and safety constraints.

B. Training Pipeline

The training pipeline is illustrated in Fig. 1.

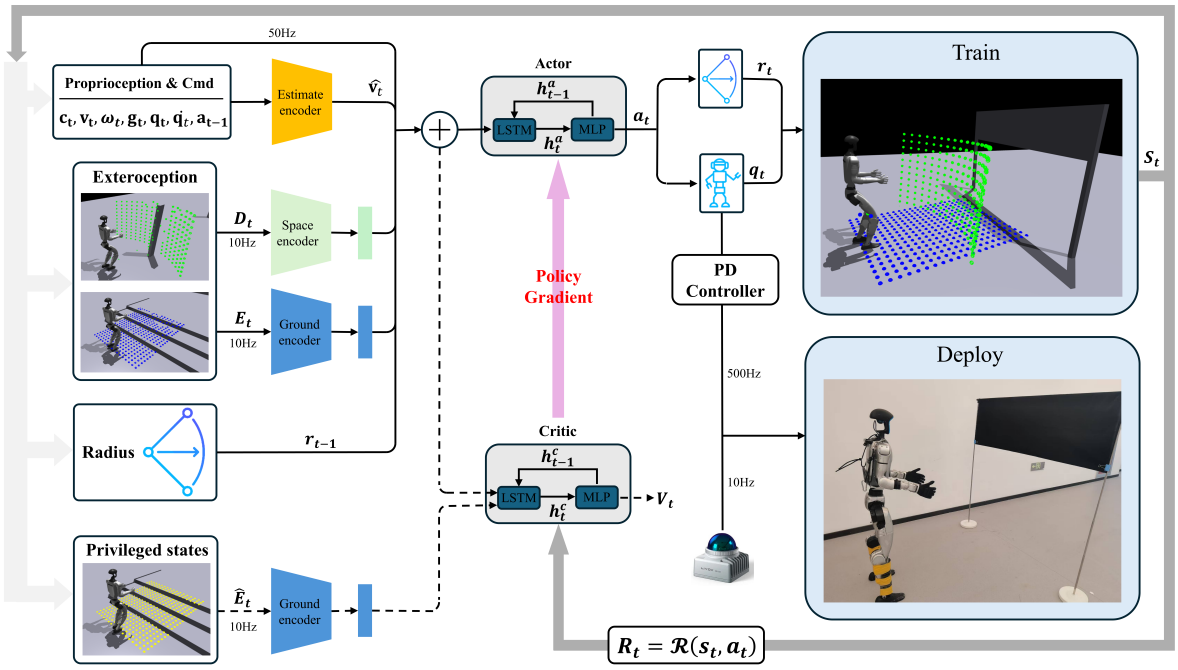

Fig. 1: Illustration of the training architecture for the proposed control framework ADAPT.

1) Observation Space: The observation vector o_t integrates proprioceptive feedback, task commands and exteroceptive data, which is defined as

$$\mathbf{o}_t = \{\mathbf{c}_t, \mathbf{v}_t, \boldsymbol{\omega}_t, \mathbf{g}_t, \mathbf{q}_t, \dot{\mathbf{q}}_t, \mathbf{a}_{t-1}, \mathbf{E}_t, \mathbf{D}_t, r_{t-1}\}. \quad (2)$$

The specific components in (2) are:

- *Task Command:* The velocity command $\mathbf{c}_t \in \mathbb{R}^3$.
- *Proprioceptive States:* These states include base linear velocity $\mathbf{v}_t$, base angular velocity $\boldsymbol{\omega}_t$, the gravitational acceleration $\mathbf{g}_t$, joint positions $\mathbf{q}_t$, joint velocities $\dot{\mathbf{q}}_t$ and the preceding action $\mathbf{a}_{t-1}$.
- *Exteroceptive States:* During learning, synthetic point clouds are acquired via the LiDAR simulation framework from Wang et al. [23]. We rasterize the raw point cloud into grid structures used by both the actor and critic to facilitate efficient processing. Consequently, this input is processed into two orthogonal robot-centric maps as shown in Fig. 1: (1) A *horizontal elevation map* ($\mathbf{E}_t \in \mathbb{R}^{21\times17}$) that captures terrain geometry via discrete height sampling. (2) A *vertical distance map* ($\mathbf{D}_t \in \mathbb{R}^{13\times17}$) that encodes spatial traversability. The distance map is generated by projecting points into a cylindrical sector spanning from -45 to $+45$ degrees around the vertical axis, where each point records the distance to the nearest obstacle.
- *Adaptive Perception Radius:* The radius r_{t-1} controls only the radial sensing range of the vertical distance map $\mathbf{D}_t$. The horizontal elevation map $\mathbf{E}_t$ uses a fixed spatial extent and sampling resolution, both independent of this radius. It is constrained to $r_t \in [1, 5]$ m throughout training and deployment. The vertical distance map uses a cylindrical sector grid with a fixed number of bins $(N_r, N_\theta) = (13, 17)$ as illustrated in Fig. 2. For a selected radius r_{t-1}, the radial sampling interval is defined as $\delta_r = r_{t-1}/N_r$, which makes the radial sampling interval of the vertical distance map grow linearly with the sensing radius; the angular span and number of bins remain fixed. Therefore, a larger radius extends the perceptual horizon but coarsens sampling resolution, while a smaller radius concentrates the grids and yields finer resolution. When long-range perception is unnecessary, reducing the sensing radius can also limit the influence of distal sensor noise by excluding distant returns from the map.

The critic operates on a privileged observation space to facilitate accurate estimation of the value function V_t. In addition to the actor's input vector o_t, the critic receives the noise-free ground-truth elevation map $\hat{\mathbf{E}}_t$ directly acquired from the simulation engine.

2) Action Space: At 50 Hz, the policy outputs $\mu_t \in \mathbb{R}^{13}$. Its first 12 elements set joint targets for a low-level Proportional-Derivative (PD) controller:

$$q_t = s_a \cdot \mu_{t,1:12} + q_{\text{default}}, \quad (3)$$

where s_a is the action scale and q_{default} the nominal configuration. *Adaptive Perception Radius:* The final element sets $r_t \in [1, 5]$ m, adjusting only the next vertical distance map's radial range and sampling interval. This empirical range avoids insufficient lookahead below 1 m and excessive distal noise above 5 m.

3) Reward Functions: The reward function $\mathcal{R}$ integrates multiple objectives to encourage robust and efficient locomotion adhering to safety constraints. The total reward $\mathcal{R}_t = \sum_i w_i r_i(t)$ is a weighted sum of the following components:

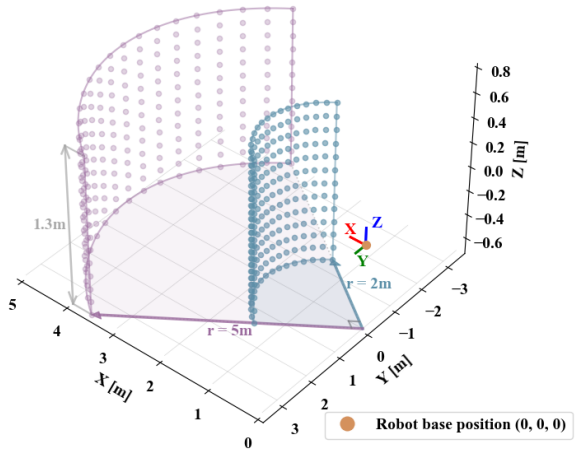

Fig. 2: Illustration of the adaptive sensing radius mechanism for the vertical distance map. The vertical distance map is constructed on a cylindrical sector with a fixed number of cells (17×13). A smaller radius (blue) provides higher spatial resolution for precise near-field sensing, while a larger radius (purple) extends the perception range at the cost of reduced resolution. The policy learns to dynamically adjust this radius based on the robot's locomotion state.

- *Motion Control:* Precise velocity tracking, postural stability, and energy efficiency are promoted by penalizing deviations in base orientation, excessive joint velocities or accelerations, and high-frequency action oscillations.
- *Safety Mechanisms:* Safe and prolonged operation is ensured by penalizing critical failures such as body collisions, mesh interpenetrations, and joint limit violations. Furthermore, survival bonuses and foot airtime rewards are provided to foster dynamic and robust obstacle clearance.
- *Active Perception:* The sensing radius of the vertical distance map is learned jointly with locomotion via the task rewards and two radius regularization terms: a smoothness term penalizes frame-to-frame radius changes, and a deviation term penalizes squared deviations from the initial radius, ensuring smooth and stable sensing adjustments.

4) Terrain Design: BarrierTrack (Fig. 3), a customized environment assembling discrete obstacle modules, is designed to cultivate versatile locomotion. Each episode features a sequence of five obstacles of the same terrain type with progressively increasing difficulty according to a curriculum schedule.

A curriculum-based strategy is adopted where each terrain type τ is parameterized by a difficulty scalar $s \in [0, 1]$, and its generation parameters are interpolated as $\mathbf{p}_\tau(s) = (1 - s)\,\mathbf{p}_\tau^{\min} + s\,\mathbf{p}_\tau^{\max}$ based on Table II. Perlin noise is superimposed onto ground surfaces to simulate terrain irregularity and prevent overfitting.

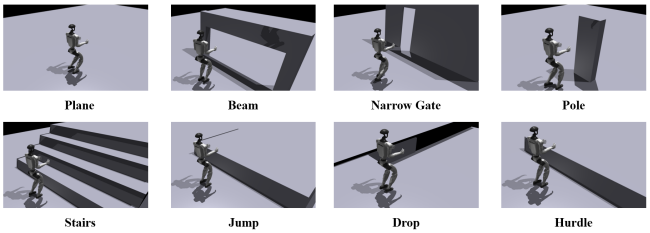

Fig. 3: Examples of procedurally generated obstacles in our *BarrierTrack* training environment. The curriculum includes a diverse set of challenges such as stairs, drop, hurdles, and constrained passages like beams and narrow gates, promoting the development of a versatile locomotion policy.

TABLE II: Parameters for generating curriculum training terrains. The ↑ symbol indicates the parameter value increases as difficulty scales up ($\mathbf{p}_\tau^{\min} < \mathbf{p}_\tau^{\max}$), while the ↓ symbol indicates the value decreases as the task becomes harder ($\mathbf{p}_\tau^{\min} > \mathbf{p}_\tau^{\max}$).

Terrain	Param. (m)	$\mathbf{p}_\tau^{\min}$	$\mathbf{p}_\tau^{\max}$
Beam	Clearance Height ↓	1.33	1.10
	Beam Depth ↑	0.20	0.20
Narrow Gate	Gate Width ↓	0.80	0.45
	Position Offset ↑	0.00	0.60
Pole	Pole Width ↓	0.50	0.10
	Position Offset ↑	0.00	0.00
Stairs	Step Height ↑	0.10	0.30
	Step Length ↓	0.50	0.30
Jump	Wall Height ↑	0.05	0.40
	Wall Depth ↑	0.10	0.30
Drop	Drop Height ↑	0.10	0.60
	Platform Length ↓	0.50	0.30
Hurdle	Hurdle Height ↑	0.05	0.40
	Hurdle Depth ↑	0.05	0.10

C. Training Architecture

As shown in Fig. 1, two separate MLP encoders map $\mathbf{E}_t$ and $\mathbf{D}_t$ to 32-dimensional features, concatenated into a 64-dimensional environmental embedding and fused with proprioception before separate actor and critic LSTMs. The critic additionally receives the noise-free simulation elevation map; the actor uses only deployable observations. Conditioned on its recurrent hidden state $\mathbf{h}_t^a$, the actor outputs joint targets and the next vertical-map sensing radius, closing the perception-control loop.

IV. Experiments

A. Experimental Configuration

1) Simulation Environment: All policies are trained using NVIDIA Isaac Gym [32] on a modified NVIDIA RTX 4090 GPU with 48 GB of VRAM, using a 12-DoF Unitree G1 humanoid robot. LiDAR point clouds are synthesized using the simulator from [23], which can accurately model sensor noises and occlusion from static terrains and dynamic robot links.

2) Real-world Environment: The trained policy is then deployed on a physical Unitree G1 robot. A Livox Mid-360 LiDAR is mounted on the head to provide raw point cloud data. State estimation and point cloud registration are handled by Fast-LIO2 [33]. The horizontal elevation map $\mathbf{E}_t$ and the vertical distance map $\mathbf{D}_t$ are extracted from the registered point cloud using the GPU-accelerated pipeline from [3] at 10 Hz. The map generation follows the exact same procedure used in simulation. Joint commands are executed at 50 Hz on an onboard NVIDIA Jetson Orin NX.

B. Simulation Experiments

1) Evaluation Setup: All methods are evaluated on the highest difficulty tier of *BarrierTrack*. The benchmark includes all seven terrain primitives, grouped into two categories: terrains that require precise foot placement like Stairs, Drop, Jump and Hurdle, and the others that require traversable-space selection like Beam, Pole and Narrow Gate. For each terrain type, every episode contains five obstacles of the same class with progressively increasing difficulty; obstacle parameters are sampled within the range given in Table II according to the curriculum schedule.

2) Metrics: Robot performance is evaluated by the mean and standard deviations over two metrics: (1) Success Rate (R_{succ}), which is the fraction of discrete obstacles that the robot successfully navigates within a single trial, i.e., the number of obstacles crossed divided by 5. and (2) Normalized Traversal Progress (R_{trav}), which is the forward distance traveled along the course normalized by the total track length. Note that R_{trav} is typically no lower than R_{succ}, as the robot may stumble forward for some distance even after failing to clear an obstacle, thereby accumulating partial traversal progress.

3) Performance Comparison: ADAPT is benchmarked against four baselines to evaluate its navigation capability and computational efficiency. Each method is evaluated across all seven terrain types under three speed commands: $v \in \{0.5, 1.0, 1.5\}$ m/s. For each terrain-speed combination, 10 trials are conducted and the average results are reported. The four selected baselines are:

- PIM [4]: A single-stage training method that uses an elevation map from LiDAR point cloud as exteroceptive input. Contrastive learning aligns the generated elevation map with ground truth.
- Parkour [14]: A two-stage method where a teacher policy is first trained with ground-truth elevation maps and then agile locomotion skills are transferred through supervised distillation to a student policy that only uses depth images.
- TSDF: A camera-based approach that performs 3D reconstruction via nvblox [34] and feeds local grid TSDF values as exteroceptive input for policy training.
- Gallant [21]: A voxel-based approach that provides full 3D awareness but at a higher computational cost.

Table III details the performance comparison, while Table IV(a) presents the associated training costs. ADAPT (Ours) achieves the highest success rates across all terrains while requiring less training time per iteration and lower peak GPU memory than the Gallant and TSDF baselines. The performance of the baselines is consistent with the sensing characteristics within each method. PIM relies on a 2.5D elevation map and shows reduced performance on Pole and Narrow Gate terrains where vertical and lateral clearance cues are important. Parkour uses depth images and performs less reliably on terrains requiring broader spatial coverage, such as Jump, Hurdle and Beam. TSDF provides volumetric awareness, but its performance remains limited on fine-grained terrains although incurring the highest training cost. Gallant improves spatial coverage relative to the other baselines, but its higher-dimensional voxel input requires higher computational cost and achieves lower success rates than ADAPT on most terrains. Overall, ADAPT provides a favorable balance between traversal performance and computation efficiency across various tasks.

Although voxels preserve richer 3D geometry, our Gallant implementation trades sensing coverage for fine resolution under a fixed voxel count and computational budget. Its performance gap widens at high speed, where obstacles entering view trigger abrupt deceleration and occasional imbalance, suggesting insufficient look-ahead for smooth motion adjustment. ADAPT instead adjusts its look-ahead distance to the robot's motion state.

All methods use identical hardware and training settings (parallel environments, rollout length, PPO epochs, and batch settings). Table I compares per-frame, pre-encoding actor exteroceptive dimensions: Gallant uses $32\times32\times40 = 40{,}960$ values and ADAPT uses $21 \times 17 + 13 \times 17 = 578$, a 98.6% reduction. Proprioception, observation history, and critic-only inputs are excluded. Table IV(a) reports wall-clock iteration time and peak single-GPU memory monitored with `nvidia-smi`. Relative to Gallant, ADAPT reduces iteration time from 34.6 to 11.3 s (32.7% of the baseline) and peak memory from $36{,}054$ to $15{,}870$ MiB (56.0% lower).

PIM has the lowest training cost but lacks body-clearance information. ADAPT incurs a moderate cost increase over PIM while remaining cheaper than the volumetric baselines.

4) Ablation Study on Adaptive Mechanism: To specifically validate the obstacle avoidance capability of the adaptive mechanism, the ablation study focuses on Beam, Pole and Narrow Gate terrains where perception is critical. For each method, terrain, and speed command, 10 trials are conducted in simulation. Table V reports the success rate averaged over the three terrains.

Besides the proposed method, two additional settings isolating the effect of radius representation and reward formulation are considered: 1) *Fixed Radius*, where the sensing radius r_t is excluded from the observation and action spaces. The policy operates with a constant radius. Separate policies are trained with $R \in \{1, 3, 5\}$ m. 2) *w/o Adaptive Reward*, where r_t remains both an observation and an action, but the `radius_smoothness` and `radius_regularization` reward terms are removed. All other reward terms are retained.

Table V summarizes the ablation results. As in the main

TABLE III: Benchmarked Comparison in Simulation. R_{succ}: success rate (%, ↑). R_{trav}: normalized traversal progress (%, ↑).

Method	Plane		Stairs		Drop		Jump	
	R_{succ}	R_{trav}	R_{succ}	R_{trav}	R_{succ}	R_{trav}	R_{succ}	R_{trav}
PIM [4]	100 (±0.0)	100 (±0.0)	65.33 (±37.57)	70.00 (±34.61)	99.33 (±3.59)	99.77 (±1.26)	90.67 (±16.92)	91.67 (±14.62)
Parkour [14]	100 (±0.0)	100 (±0.0)	28.00 (±28.10)	37.20 (±24.93)	91.33 (±14.31)	92.40 (±13.52)	44.67 (±20.45)	51.40 (±17.49)
TSDF	100 (±0.0)	100 (±0.0)	40.00 (±20.00)	47.43 (±17.50)	56.67 (±24.81)	63.33 (±22.75)	43.33 (±13.74)	48.63 (±13.65)
Gallant [21]	100 (±0.0)	100 (±0.0)	70.00 (±22.95)	76.30 (±21.75)	80.00 (±31.41)	83.23 (±27.62)	68.00 (±13.27)	74.10 (±11.34)
Ours	**100** (±0.0)	**100** (±0.0)	**94.67** (±13.60)	**95.90** (±10.63)	**100.00** (±0.00)	**100.00** (±0.00)	**98.67** (±4.99)	**98.90** (±4.12)

Method	Hurdle		Beam		Pole		Narrow Gate	
	R_{succ}	R_{trav}	R_{succ}	R_{trav}	R_{succ}	R_{trav}	R_{succ}	R_{trav}
PIM [4]	82.00 (±20.88)	86.37 (±16.44)	66.67 (±9.43)	75.53 (±5.28)	2.67 (±6.80)	6.17 (±7.09)	27.33 (±15.04)	31.17 (±14.17)
Parkour [14]	47.33 (±13.15)	52.67 (±12.81)	14.00 (±11.72)	20.53 (±6.93)	5.33 (±8.84)	15.00 (±7.90)	48.00 (±16.81)	53.03 (±16.05)
TSDF	49.33 (±11.23)	55.47 (±13.98)	33.33 (±11.93)	40.83 (±11.37)	7.33 (±9.64)	14.37 (±9.07)	22.67 (±16.11)	29.80 (±14.31)
Gallant [21]	54.67 (±16.28)	62.47 (±14.21)	32.00 (±25.61)	40.40 (±27.17)	46.00 (±44.17)	49.40 (±41.82)	66.00 (±25.90)	69.57 (±23.35)
Ours	**90.00** (±14.38)	**91.80** (±12.80)	**96.67** (±7.45)	**97.57** (±5.47)	**98.00** (±6.00)	**98.57** (±4.79)	**80.00** (±18.62)	**83.03** (±16.30)

TABLE IV: Training and Runtime Efficiency. Runtime latencies: averages of per-trial means/P95s (ms). ADAPT reports the larger of the two map-stream statistics for each map-latency and perception-to-publish latency entry. Resources: mean whole-host CPU utilization/RAM usage and maximum sampled VRAM usage.

(a) Training efficiency

Method	Train/iter (s)	Peak VRAM (MiB)
PIM [4]	2.6	10485
Parkour [14]	97.8	28791
Gallant [21]	34.6	36054
TSDF	246	38473
ADAPT (Ours)	**11.3**	**15870**

(b) Local-computer runtime latency and resource usage

Method	Map latency Mean/P95 (ms)	Inference latency Mean/P95 (ms)	Perception-to-publish latency Mean/P95 (ms)	Peak VRAM (MiB)	CPU / RAM (% / GiB)
Gallant	64.06 / 87.49	1.91 / 3.20	79.45 / 106.09	5711	38.19 / 10.66
ADAPT	**41.64 / 64.09**	**1.40 / 1.96**	**55.94 / 79.31**	**2373**	**17.76 / 8.04**

benchmark, each trial evaluates traversal through a sequence of five obstacles of the same terrain type at a fixed speed command. The success rate R_{succ} is the fraction of the five obstacles successfully crossed in a trial. The table reports the mean and standard deviation over the 30 trial-level success rates pooled across Beam, Pole, and Narrow Gate at each speed.

A short fixed radius ($R = 1$) yields the lowest average success rates, particularly at higher speeds, indicating that the limited look-ahead range is insufficient for these terrains. A long fixed radius ($R = 5$) performs better than $R = 1$ at high speeds but remains inferior to a moderate fixed radius ($R = 3$), suggesting a trade-off between larger look-ahead range and reduced local resolution. However, the performance of a fixed radius is still behind the fully adaptive method. Removing both radius-related rewards in *w/o Adaptive Reward* lowers success at higher speeds, where less time for perception and avoidance may increase sensitivity to unsuitable or fluctuating sensing ranges. Together, the smoothness and regularization terms help stabilize radius selection and its coordination with locomotion. Overall, the fully adaptive method achieves the highest average success rate, indicating efficacy of the proposed closed-loop radius modulation.

TABLE V: Ablation on adaptive sensing radius. R_{succ}: success rate (%). Values are averaged over Beam, Pole, and Narrow Gate terrains.

Type	Config	v=0.5 m/s	v=1.0 m/s	v=1.5 m/s
Fixed Radius	$R = 1$	51.33 (±17.65)	51.33 (±23.49)	24.67 (±22.32)
	$R = 3$	81.33 (±21.25)	72.67 (±15.90)	56.67 (±29.70)
	$R = 5$	64.00 (±18.90)	64.67 (±26.17)	53.33 (±30.26)
w/o Adap. Reward	no r_{adapt}	92.67 (±10.93)	71.33 (±25.66)	42.67 (±29.09)
Adaptive Radius	**Ours**	**94.67** (±10.24)	**96.00** (±8.00)	**84.00** (±19.60)

C. Real-world Experiments

1) Deployment: ADAPT transfers directly to the Unitree G1 without fine-tuning (Fig. 4). For Fig. 5, final checkpoints from ten independent training seeds are evaluated in matched simulation and real-world scenes with a fixed single-obstacle

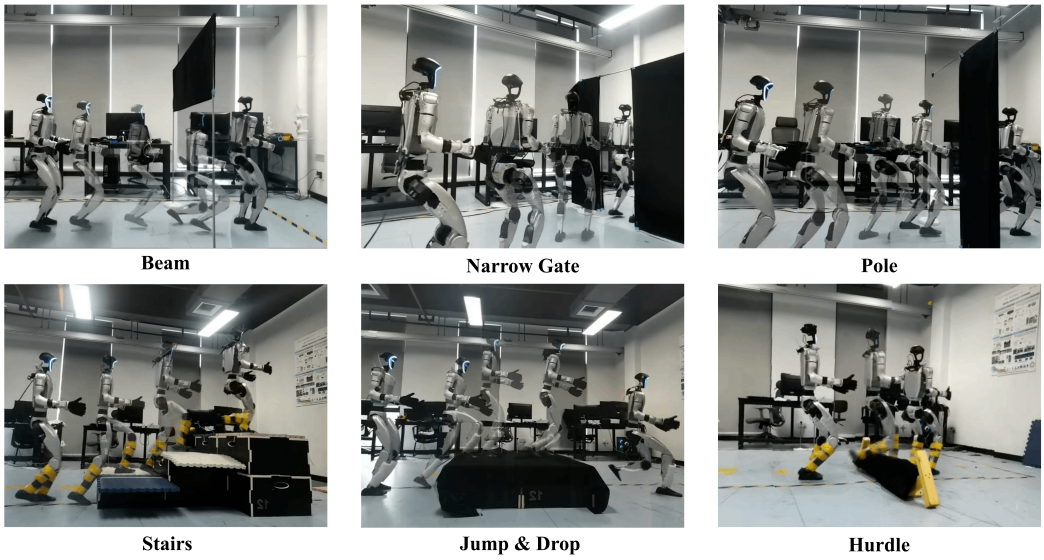

Fig. 4: Zero-shot real-world deployment across diverse terrains. The trained policy enables the physical robot to successfully traverse seven distinct *BarrierTrack* terrains without any real-world fine-tuning.

configuration per terrain. Each policy is tested once per terrain at each speed ($v \in \{0.5, 1.0, 1.5\}$ m/s), totaling 30 trials per terrain and domain.

In terrains that require traversable-space selection, the robot identifies viable regions using the vertical distance map $\mathbf{D}_t$. When encountering aerial obstacles such as beams, the robot lowers its body to crouch underneath the beams and resumes to the normal posture afterward. For narrow pillar-like obstacles, the robot autonomously adjusts its heading to bypass the poles. When facing narrow gates, it precisely aligns its orientation with the traversable path. In terrains requiring precise foot placement, the robot relies on the elevation map $\mathbf{E}_t$ to stably ascend stairs, negotiate drops and jumps, and step over hurdles. The adaptive sensing radius of the vertical distance map further enables the policy to regulate its perception horizon online during high-speed locomotion.

2) Results: Fig. 4 demonstrates zero-shot sim-to-real transfer across all seven terrains, while Fig. 5 presents the corresponding per-terrain success rates in simulation and physical deployment. Despite an inherent sim-to-real domain gap, the real-world success rates closely follow the trend of their counterparts in simulation, maintaining at least 80% success rates across all terrains. The physical robot exhibits a range of whole-body behaviors, including stair ascent and descent, stepping across drops and gaps, crouching under beams of varying height, and navigating through narrow passages and pole fields, suggesting good generality and practical applicability of the learned policy.

3) Runtime Efficiency: Table IV(b) compares ADAPT and Gallant on a local computer (i7-12700H, RTX 3060 Laptop GPU with 6 GiB VRAM). These runtime benchmarks are performed on the local computer, not on the onboard Jetson Orin NX. Each method is tested five times on each of three

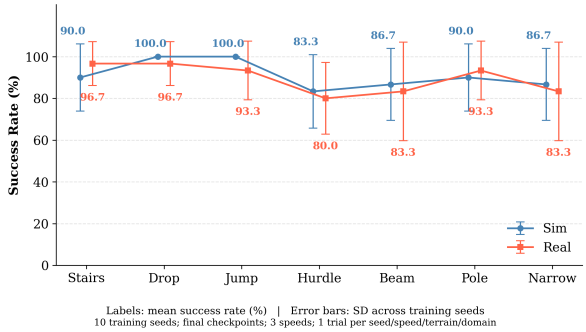

Fig. 5: Single-obstacle traversal success rates in matched simulation and real-world scenes. Points and error bars show the mean and sample standard deviation across ten training seeds, after averaging each policy's outcomes over three speeds.

terrain categories (flat, single obstacle, and complex environment). Map latency spans the input callback to the first observed map-ready event; perception-to-publish latency spans the same input callback to the first command-publication call using that update. Inference latency is measured using current-stream CUDA events. ADAPT reduces mean perception-to-publish latency by 29.6% using the larger of the two stream means and peak GPU memory by 58.4% relative to Gallant, with lower CPU/RAM usage.

V. Conclusions

This paper presented ADAPT, an adaptive dual-projection framework for perceptive humanoid traversal in complex 3D environments. By decomposing local geometry into a horizontal elevation map and a vertical distance map, ADAPT explicitly separates ground-support and body-level clearance

constraints into a compact representation for high-frequency control. Furthermore, embedding the sensing radius into the action space enables the robot to dynamically adjust its perceptual horizon based on motion context. Experiments demonstrate that ADAPT significantly improves traversal robustness and computational efficiency over fixed-range and voxel-based alternatives, achieving successful zero-shot sim-to-real transfer on a Unitree G1 robot.

Despite these capabilities, the current framework focuses solely on geometric traversability and adapts only the sensing radius. Future work will investigate richer active perception policies that jointly optimize sensing scale, resolution, and update frequency, while incorporating semantic reasoning to enable long-horizon navigation in complex, human-centered environments.

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